



DESCRIPTION OF COURSEWORK

Course Code	MAT315
Course Name	Time Series
Lecturer	Koh Siew Khew
Academic Session	September 2025
Assessment Title	Time Series Project

A. Introduction/ Situation/ Background Information

This is a group project, with two students in one group. The objective is to apply time series techniques and concepts learned in the course to analyse the provided data. The project involves exploring and describing the data, checking for stationarity, fitting time series regression model and identifying suitable models using tools such as the autocorrelation function (ACF) and partial autocorrelation function (PACF). You will fit and compare AR, MA, ARMA, and ARIMA models, and generate forecasts. All modeling choices and results should be clearly justified. You are expected to demonstrate your understanding of time series modeling by drawing accurate conclusions and providing well-supported interpretations based on your analysis.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

Perform time series analysis using the statistical software R.

C. University Policy on Academic Misconduct

Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

- i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
- ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
- iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use

materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

- Form a group of two persons and fill up the form: [MAT315 Project_Sep 2025.xlsx](#) by **Friday, 5 December 2025**. (The link to the form is available on Moodle)
- You are required to submit 3 files (A folder will be share with you when the deadline is near):
 1. A .txt file that consists of your R commands (just the codes, no output):
[ABC1234567_XYZ1234567.txt](#)
 2. Cover Page combines with a .pdf file that consists of your answers to the questions:
[ABC1234567_XYZ1234567.pdf](#)
Use the XMUM assignment template (word document) made available to you. Start your answer from page 3. You may choose to type your answers. R output must be included. Convert the word document into a pdf file before submitting it.
 3. A presentation video. About 10 minutes (each student 5 minutes) for two members.
- Limit your report to a maximum of 30 pages, with an ideal length of 15 to 20 pages.
- Only one group member should submit the report and presentation video via XMDrive. Do not submit multiple copies - only one submission per group is required. A submission link will be provided closer to the due date.
- You **do not need** to attach the marking rubrics.
- At the end of the report, list down the contributions from each group member.
- Due date to submit report and presentation video: **Friday, 26 December 2025**.

E. Data Description

The Department of Agriculture monitors monthly rice yield index to maintain national food security. You are provided with a dataset, “rice_yield.csv”, containing monthly observations. Your task is to analyze the historical data to understand the key characteristics of the series, verify its statistical properties, and develop a suitable forecasting model. In the csv file, x represents the rice yield index (YI) which is a standardized measure of average rice productivity over time.

F. Task

- *Time Plot & Description*
Produce a time plot of the rice yield index. Based on the plot, describe the overall trend and discuss any seasonal patterns observed in the series. Then, construct a boxplot (by month) to examine the seasonal variation (if any) in the data.
- *Time Series Decomposition*
Apply a suitable decomposition model to the series, and justify which model is more appropriate for this dataset.
- *Correlation Structure*
Plot the correlogram of the series. From the plots, comment on whether the series appears to be stationary or non-stationary.
- *Fitting a Time Series Regression Model*
Fit a suitable time series regression model to the data (e.g., including a trend and/or seasonal components if appropriate). Compare the fitted and observed values, and comment on how well the model explains the data.
- *Fit and Compare Models*
Fit one or more suitable models from among AR, MA, ARMA, ARIMA or SARIMA to the series. Justify your models selection based on the series’ characteristics and the ACF/PACF patterns. Compare the fitted models using appropriate criteria (e.g., AIC or residual diagnostics) and identify the best-fitting model.
- *Forecasting*
Using the selected best-fitting model, produce forecasts for the next 12 months. Display the forecasts with 95% confidence intervals and provide an interpretation of the results.
- *Volatility Check*
Plot the squared residuals of the best-fitting model. Comment on whether there is evidence of volatility clustering, and discuss if an ARCH or GARCH model might be appropriate for the residual series.

APPENDIX 1

MARKING RUBRICS

Report and Analysis: 80%

(Formatting of the report is also assessed within each section under “Criteria.”)

Component Title	Report			Total Marks		80	
Criteria	Score (s)					Weightage (w)	Marks (s × w)
	Excellent	Good	Average	Need Improvement	Poor		
Time Plot & Description	5	4	3	2	1	1.5	
Time Series Decomposition	5	4	3	2	1	1.5	
Correlation Structure	5	4	3	2	1	1.5	
Fitting a Time Series Regression Model	5	4	3	2	1	4	
Fit and Compare Models	5	4	3	2	1	4	
Forecasting	5	4	3	2	1	2	
Volatility Check	5	4	3	2	1	1.5	
	TOTAL						

Presentation: 20%

Component Title	Report			Total Marks		20	
Criteria	Score (s)					Weightage (w)	Marks (s × w)
	Excellent	Good	Average	Need Improvement	Poor		
Appearance	5	4	3	2	1	0.5	
Content	5	4	3	2	1	0.5	
Delivery Skills	5	4	3	2	1	1.5	
Clarity of Explanation	5	4	3	2	1	1.5	
	TOTAL						